Open-Set Domain Adaptation via Adaptive Separation and Two-Level Alignment With Application in SAR Target Recognition

Yuchen Guo, Bin Yang, Lan Du, *Senior Member, IEEE*, and Jian Chen

***Abstract*—Open-set domain adaptation (OSDA) recognition assumes distribution shifts between labeled source domain data and unlabeled target domain data, with the target domain containing additional unknown class. The goal of OSDA recognition is to classify the known classes and identify the unknown class in target domain. Typical OSDA recognition studies first separate known and unknown classes in the target domain, then align shared known classes between source and target domains, but still face the issues of insufficient separation and negative alignment due to the confusion between unknown and known classes in the target domain. To address these challenges, this article proposes an OSDA recognition method via adaptive separation and two-level alignment (ASTA), and applies it to synthetic aperture radar target recognition. In the separation phase, to achieve precise separation of known and unknown class targets in the target domain and minimize the influence of unknown targets for knowledge transfer in the alignment phase, the proposed adaptive separation module constructs dynamically adjusted separation thresholds for different batches. In the alignment phase, to accurately transfer the classification knowledge from the source domain, this article introduces the multilevel alignment framework from closed-set domain adaptation, achieving cross-domain knowledge transfer at both domain level and class level. In particular, to ensure precise transfer of known class classification knowledge at the class-level alignment, the distance between cross-domain prototypes (constructed by prototype network) of the same class is minimized. To mitigate the negative impact of unknown class targets (missed during the separation phase) on target domain prototype construction, a prototype-based pseudolabel filtering strategy and a prototype construction loss weighting mechanism are proposed. The OSDA recognition results on the SAMPLE and Office-31 datasets demonstrate that ASTA achieves SOTA performance, proving its superiority.**

***Index Terms*—Adaptive separation (AS), multilevel alignment, open-set domain adaptation (OSDA), prototype network, synthetic aperture radar (SAR), target recognition.**

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I. Introduction

Being fundamental to computer vision systems, target recognition holds substantial practical value in diverse fields from transportation automation to military security [1], [2]. For example, in the field of military security, target recognition can quickly identify and classify potential threats from the enemy, providing commanders with detailed information to make timely decisions and resource deployments. Contemporary computer vision systems have undergone paradigm shifts through deep neural architectures, achieving remarkable breakthroughs in target recognition tasks when leveraging large-scale annotated data repositories. However, this data-driven paradigm introduces critical bottlenecks in model deployment due to its inherent dependence on meticulously labeled training corpora, particularly evident in domain adaptation scenarios requiring cross-environment generalization. For example, in synthetic aperture radar (SAR) target recognition, the time-consuming and labor-intensive nature of labeling measured SAR image has resulted in insufficient training data, which has impacted the training and application of SAR target recognition models. Utilizing labeled data from other domains to assist in training models to classify target data is a feasible approach, such as using labeled simulated SAR data to aid in training SAR target recognition models for measured SAR data. However, cross-domain transfer scenarios present fundamental challenges, as domain-shift phenomena arising from statistical distribution discrepancies between source and target domains severely compromise the direct applicability of auxiliary domain-optimized models. Unsupervised domain adaptation (UDA) is a feasible and effective solution [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13]. UDA mitigates cross-domain divergence via feature space harmonization across aligned source and target domain observations, optimizing cross-domain information flow for enhanced knowledge transfer efficiency.

However, existing UDA follows the closed-set assumption, which means that the source domain and the target domain share the same class space, and this is unreasonable. A more realistic assumption is that the target domain may present additional new categories that need to be recognized as a unified unknown class. For instance, in battlefield scenarios, adversaries may deploy novel military targets (e.g., new types of armored vehicles or autonomous aerial vehicles). Due to the lack of simulation modeling data for these new targets, existing unsupervised closed-set domain adaptation (CSDA) methods exhibit constrained applicability in open-set scenarios, where indiscriminate domain-invariant representation (DIR) learning under cross-domain distribution alignment risks inducing detrimental knowledge transfer stemming from latent semantic inconsistency, resulting in the failure of CSDA to identify new targets. Open-set recognition methods provide insights for identifying new targets. By incorporating the ideas of open-set recognition to extend CSDA recognition to unsupervised open-set domain adaptation (OSDA) recognition, new targets can be directly identified without reliance on additional intelligence and manual intervention. Nowadays, some scholars have proposed OSDA recognition methods [14], [15], [16], [17], [18], [19], [20], [21], [22]. The earliest concept of OSDA recognition was initially proposed by Busto et al. [14], who assumed that both the source and target domains have shared classes (known classes) and unknown class. Subsequently, Saito et al. [15] established a paradigm where novel category instances are exclusively situated in the target distribution, a framework that has become foundational in subsequent OSDA investigations. Saito et al. [15] separated known and unknown classes in the target domain using a fixed threshold, then aligned the known classes through adversarial domain adaptation while keeping the unknown classes in their original feature space. Liu et al. [16] developed a hierarchical refinement framework employing dual discriminative operations to enable feature space differentiation between established knowledge clusters and emergent category representations within cross-domain adaptation scenarios. Silvia et al. [17] pioneered the integration of self-supervised representation learning into OSDA recognition, employing an enhanced rotation-based task to simultaneously achieve domain alignment and known–unknown class discrimination. Wang et al. [19] built on previous work [23] by exploring the clustering structure information in the target domain, gradually expanding the classification space for known and unknown classes to label target samples. Li et al. [20] considered pixel block granularity knowledge based on Busto et al.'s work, achieving better results through unbiased source domain learning and domain alignment.

The aforementioned OSDA recognition approaches follow a "separating-then-alignment" solving strategy, which means first separating the known and unknown classes in the target domain, then aligning the known classes across domains. However, in the separation phase, existing OSDA recognition methods do not perform well enough to distinguish between known and unknown classes in the target domain, which can significantly negatively impact the knowledge transfer of known classes during the alignment phase. In the alignment phase, existing OSDA recognition methods suffer from the negative alignment problem: first, existing OSDA recognition methods exhibit inaccurate transfer of cross-domain known class classification knowledge, potentially leading to incorrect alignment between known classes; second, the negative influence of unknown class targets that might have been missed during the separation phase on the knowledge transfer of known classes is not considered, which could result in incorrect alignment between known and unknown classes.

Thus, this article follows the "separation-then-alignment" solving strategy, designing an OSDA recognition method called adaptive separation and two-level alignment (ASTA), which addresses the shortcomings of existing methods and has been validated on SAR target recognition dataset. First, to accurately separate known and unknown classes in the target domain and minimize the negative impact of unknown class targets on knowledge transfer, ASTA proposes an adaptive separation (AS) module. The AS module calculates AS thresholds for each batch of target domain samples by leveraging the distance relationship between the classifier's final layer weight matrix and the target domain samples. The threshold is then used to separate unknown classes from known classes in the target domain. Then, to address the negative alignment issue in the alignment phase, ASTA introduces for the first time the multilevel alignment strategy from CSDA into OSDA recognition, achieving cross-domain known class knowledge transfer at both the domain level and the class level. Considering that unknown class targets may still exist in the target domain, we have adapted the class-level alignment to make it suitable for OSDA recognition. Specifically, during class-level alignment, we propose a prototype-based pseudolabel filtering strategy and a prototype construction loss weighting mechanism to eliminate the negative impact of unknown class targets (missed during the separation phase) on target domain prototype construction while improving the accuracy of known class prototype construction in the target domain.

This work makes the following methodological contributions.

1) The proposed AS module dynamically constructs separation thresholds for each batch of data, achieving precise known–unknown target separation in the target domain. This effectively eliminates the influence of the unknown class to the greatest extent for knowledge transfer during the alignment phase.
2) The multilevel alignment strategy for CSDA was first introduced to address OSDA recognition problems, with an adapted class-level alignment to make it suitable for OSDA recognition. Specially, in the class-level alignment, the proposed pseudolabel filtering strategy and the prototype construction loss weighting not only mitigate the impact of unknown class targets (missed during the separation phase) on the construction of known class prototypes in the target domain, but also enhance the accuracy of the constructed prototypes.

II. Related Work

This section reviews prior research relevant to the ASTA method for OSDA recognition developed in this study.

A. Prototype Network

Prototype network is based on the idea of embedding, generating corresponding prototypes through clustering data points of each class in the embedding space [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]. The generated prototypes can effectively represent the structure of the entire class. To obtain the prototypes for each class and use them in classification tasks, Snell et al. [30] pioneered a neural network framework to model the nonlinear mapping between raw inputs and embedding space, subsequently deriving prototype vectors by computing featurewise mean values of sample categories within the learned embedding space. In a mathematical framework, given a sample set $S = \{X_s^i, Y_s^i\}_{i=1}^N$ containing N samples across K classes, where X_s^i is the feature vector extracted by the feature extractor and Y_s^i is the class label, the K prototypes $P_k \in \{1, 2, \ldots K\}$ are first learned through the embedding function $f_\phi(\cdot)$ in the embedding space

$$P_k = \frac{1}{|S_k|} \sum_{(X_s^i, Y_s^i) \in S_k} f_\phi(X_s^i). \tag{1}$$

Then, given a distance function $d(\cdot)$, for the sample X_s, the distance to each prototype in the embedding space is calculated, and the probabilities of the sample belonging to each class are normalized using

$$P_\phi(y = k | X_s^i) = \frac{\exp(-d(f_\phi(X_s^i), P_k))}{\sum_{i=1}^K \exp(-d(f_\phi(X_s^i), P_i))}. \tag{2}$$

Next, the cross-entropy loss between the predicted classes and the class labels is calculated, and backpropagation is performed to update the network, thereby training the prototype network to obtain the corresponding prototypes. This article leverages the ability of prototypes to effectively represent class structures by constructing known class prototypes for both simulated SAR data domain and measured SAR data domain through the prototype network, and then, minimizes the Euclidean distance between the same class prototypes across domains to facilitate more accurate class-level knowledge transfer.

B. Adversarial Domain Adaptation

In machine learning, if there is a different distribution between the training data and the real data, applying a model trained with supervision on the training data to the real data will lead to a substantial decline in performance, which is formally characterized as the domain shift phenomenon. Domain adaptation is a powerful approach to address domain shift problem. Substantial research has been conducted on domain adaptation, focusing mainly on three areas: divergence-based domain adaptation [36], [37], [38], [44], reconstruction-based domain adaptation [39], [40], [45], [46], [47], and adversarial domain adaptation [41], [42], [43], [48], [49], [50], [56], [57]. In recent years, the field of domain adaptation has witnessed numerous cutting-edge studies that have significantly advanced and expanded the discipline by addressing key challenges such as feature alignment, source-free adaptation, and multisource domain adaptation [53], [54], [55]. This article adopts the adversarial domain adaptation method. Yaroslav et al. [41] drew inspiration from the concept of generative adversarial network [51], treating the feature extractor as the generator and the domain classifier as the discriminator. They paradigm innovatively integrates gradient reversal mechanism (GRL) to establish a dynamic adversarial learning system between the feature extraction network and domain discriminator, coercively guiding the feature extractor

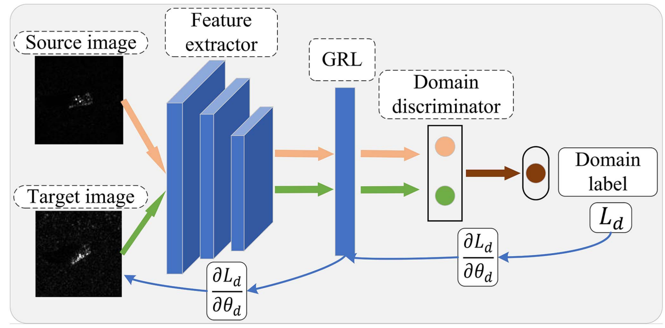

Fig. 1. Adversarial domain adaptation using GRL. The domain discriminator essentially functions as a binary classifier (predicting whether features come from the source domain or the target domain), where L_d is its loss function and θ_d represents the parameters of the model that need to be updated.

to produce domain-invariant features (DIR) with cross-domain robustness.

Specifically, as illustrated in Fig. 1, during the forward propagation process, GRL keeps the features unchanged, performing an identity transformation. In the backward propagation process, GRL reverses the gradient signs, creating adversarial dynamics where the domain classifier's gradient vectors are diametrically opposed to those computed from the feature extractor

$$L_d = -\frac{1}{n} \sum_{i=1}^{n} (y_i \log \widehat{y}_i + (1 - y_i) \log(1 - \widehat{y}_i)). \tag{3}$$

Here, $\widehat{y}_i$ is the predicted output of the domain discriminator, and $y_i \in \{0 \text{ or } 1\}$ is the domain label of the sample.

III. Method

This segment delineates a detailed introduction to the overall framework of the ASTA, the AS module, and the two-level alignment (TA) module.

A. Overall Framework

In OSDA recognition, the model has labeled source domain data $\{X_s^i, Y_s^i\}_{i=1}^{N_s}$ and unlabeled target domain data $\{X_t^i\}_{i=1}^{N_t}$, with a distribution difference between the source and target domain data. The K classes shared across both source and target domains are formally designated as known classes, while target-exclusive classes are categorized as a unified unknown class. The dual objective of the OSDA recognition framework lies in achieving precise classification of known classes within the target domain while ensuring effective recognition of novel unknown instances.

The overall framework of the ASTA is illustrated in Fig. 2. First, the AS module is developed to enable the differentiation of both known and unknown data within the target domain. Then, the TA module facilitates cross-domain knowledge transfer at two levels. At the domain-level, the unbiased domain alignment of the original model ANNA is retained, but the loss for aligning unknown class pixel blocks from the source to the target domain is removed. At the class-level, the proposed prototypical class

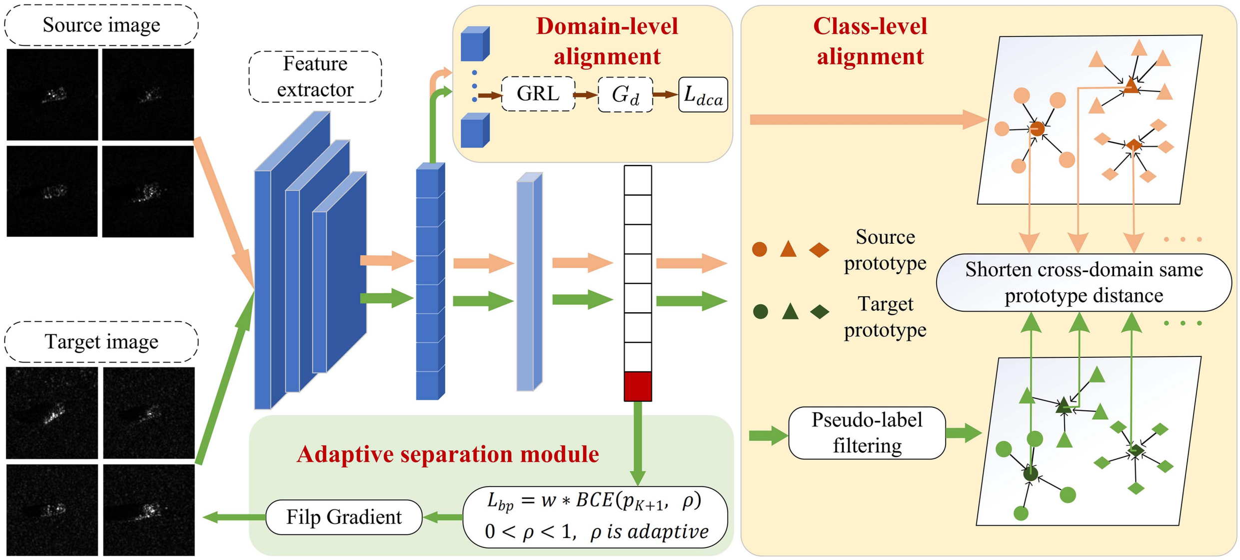

Fig. 2. Schematic diagram of the proposed ASTA architecture. In the AS module, the red box represents the classifier's prediction of the unknown class for target samples, denoted as $p_c(y = K+1|X_t)$, where ρ is the adaptive threshold and ω is the separation loss weight for each sample. In the domain-level alignment submodule, G_d refers to the dual-head domain discriminator, which can simultaneously output scores indicating whether features belong to known or unknown classes. In the class-level alignment submodule, the upper and lower quadrilateral regions represent the prototype embedding spaces of two domains, respectively. The orange circles, triangles, and diamonds represent different classes of data in the source domain, and the dark orange data points illustrate the learned prototypes of known classes in the source domain, with the green data points below representing the same concept for the target domain.

alignment submodule constructs cross-domain same category prototype relationships to facilitate accurate class alignment.

B. AS Module

The AS module is designed to establish a well-defined decision boundary between known and unknown classes within the target domain. To initiate the training process, we utilize the cross-entropy loss function to instruct the network about object categories within the source domain as follows:

$$L_s = -\frac{1}{N_s}\sum_{i=1}^{N_s} Y_s^i \cdot \log(p_c(y = Y_s^i|X_s^i)). \tag{4}$$

where $p_c(y = Y_s^i|X_s^i)$ represents the classifier's predicted score for the label class of source domain sample X_s^i.

The model must first construct an initial pseudodecision boundary for the unknown class. Therefore, the model first outputs the unknown class score for target samples through a weakly trained classifier. Specifically, the weekly trained classifier outputs the unknown class score $p_c(y = K+1|X_t) = \rho$, where $0 < \rho < 1$ represents the adaptive threshold for each batch of target samples. Next, the model employs a GRL to train the feature extractor, aiming to deceive the classifier and maximize its error rate. Furthermore, ρ is the label for the adversarial loss between the feature extractor and the classifier, meaning that ρ controls the classification of the target samples as known or unknown. Since the model reads data in batches during training, ρ should dynamically change according to different batch data.

By introducing two classifier metrics, the model dynamically adjusts the threshold ρ. The first metric m_1 collects the minimum distance between the classifier weights ω_k each time a batch of data is input into the model, where ω_k refers to the rows of the output matrix of the last layer of the classifier, with each row representing the center of a known class. $D(\omega_k)$ denotes the shortest distance between the center of the kth class and that of any other known class (where $k = 1, 2, \ldots, K$). This is determined by finding the maximum cosine similarity between the kth class center and other known class centers, which is treated as the shortest distance $D(\omega_k)$. m_1 reflects the real-time minimum distance between the centers of different known classes

$$m_1 = \frac{1}{K}\sum_{k=1}^{K} D(\omega_k). \tag{5}$$

$$m_2 = \frac{1}{K}\sum_{k=1}^{K}\left(-\frac{1}{N_{tp}^k}\sum_{j=1}^{N_{tp}^k} Y_t^j \cdot \log(p_c(X_t^j))\right). \tag{6}$$

The second metric m_2 calculates the weak classification loss for each batch of target samples, which serves to evaluate in real time the proximity of each class sample to its corresponding known class center. Specifically, based on the predictions made by the classifier for target domain samples, a fixed threshold T_1 is established to select target domain known class samples with a certain level of confidence and assign pseudolabels to these samples. By calculating the cross-entropy loss for these samples, the model can assess their proximity to the corresponding class centers, thereby obtaining the distance distribution characteristics of the current batch of target domain known-class samples. Here, X_t^j represents the target domain samples in the current batch that are assigned pseudolabels of known classes, Y_t^j denotes the pseudolabels assigned to X_t^j, and N_{tp}^k indicates the number of pseudolabeled samples belonging to the kth known class in the current batch. The impact of the hyperparameter T_1 on the model will be analyzed in the experimental section.

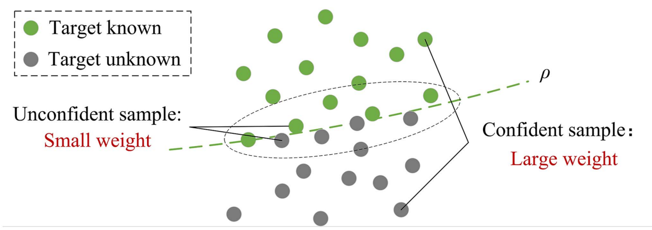

Fig. 3. Illustration of AS loss weighting. ρ is the known–unknown separation threshold. The distance of features from ρ indicates the model's confidence in classifying the feature as known or unknown; samples closer to ρ are considered uncertain and should have reduced weights, while samples further away from ρ are deemed confident and should have increased weights.

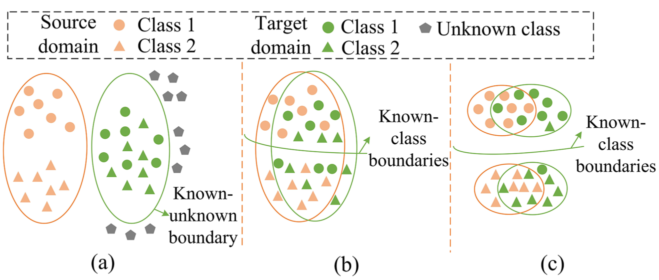

Fig. 4. Schematic illustration of domain-level alignment and class-level alignment. (a) After separating known and unknown classes in the target domain, the distributions remain unaligned. (b) Domain-level alignment aims to reduce the distance between the overall data distributions of the two domains. (c) Building upon domain-level alignment, the derived pseudoannotations-driven class-level semantic alignment achieves fine-grained category feature synchronization.

If m_1 is large, it indicates that the class centers are distributed compactly in the classification space. If m_2 is small at this time, it suggests that the known class samples of the target samples in this batch can be closely distributed around the class centers, resulting in a compact distribution of known classes; therefore, a higher threshold should be set for this batch of target samples. Conversely, if m_2 is large, it indicates that the known class samples in this batch are more sparsely distributed around the class centers, the threshold should be smaller than when it is small. The analysis holds similarly when m_1 is small. Therefore, utilizing m_1 and m_2, the model can construct a dynamic threshold ρ for each batch.

$$\rho = 1 + \ln\left(\frac{\alpha \cdot m_1}{m_2}\right). \tag{7}$$

α is an adjustable parameter whose specific selection depends on the dataset. In this article, α is set to 3.5.

The adaptive threshold ρ has already been able to accurately delineate the known–unknown separation boundary for each batch of target samples, but there is still room for further refinement. As shown in Fig. 3, compared to confident samples with $p_c(y = K + 1|X_t)$ that are far from ρ, those uncertain samples with $p_c(y = K + 1|X_t)$ lingering near the threshold ρ are less likely to accurately represent the known–unknown attribution of the samples. Therefore, we aim to reduce the weight of uncertain samples in the known–unknown separation loss while increasing the weight of confident samples. For example, if the threshold

Algorithm 1: The Execution Flow of the AS Module.

Input: Features $\{X_t^i\}_{i=1}^{N_t'}$ of each batch of target domain samples extracted by the feature extractor, where N_t' denotes the number of target domain samples in the current batch; The total number of batches per epoch N_b.
Output: Optimized the model's separation of known and unknown classes in the target domain.
1: **for** batch $= 1, 2, \ldots, N_b$ **do**
2: Calculate the m_1 and m_2 for the current batch;
3: Based on m_1 and m_2, compute ρ for the batch;
4: Calculate the ω for each X_t^i in current batch;
5: Calculate the separation loss L_{bp} for the current batch;
6: Backpropagate L_{bp} with gradient reversal layer.
7: **end for**

ρ for a batch is calculated to be 0.4, and one target sample in that batch has an unknown class score of 0.95, while another has a score of 0.41, it is evident that the model is very confident about the prediction for the first sample, and its prediction is likely correct. Therefore, the weight of this sample should be increased in the known–unknown boundary classification loss. Conversely, although the second sample's unknown class score exceeds the value of ρ, the model lacks confidence in its prediction, and we cannot guarantee the correctness of that prediction. Thus, the weight of this sample should be decreased in the known–unknown boundary classification loss. Therefore, based on the unknown class scores of different samples in the batch, we have designed corresponding known–unknown separation loss weights for each target sample, which are used for weighting the separation loss. The separation loss function weighted by the separation weights ω is shown in (9).

$$\omega = \begin{cases} \frac{p_c(y=K+1|X_t)-\rho}{1-\rho}, p_c(y = K + 1|X_t) > \rho \\ \\ \frac{\rho-p_c(y=K+1|X_t)}{\rho}, p_c(y = K + 1|X_t) \leq \rho. \end{cases} \tag{8}$$

$$L_{\text{bp}} = \omega \cdot \text{BCE}(\, p_c(y = K + 1|X_t),\ \rho) \tag{9}$$

where BCE$(\cdot)$ refers to the binary cross-entropy function. By employing a weighted separation constraint, the model can better achieve the known–unknown separation of target domain samples, leading to a more accurate delineation of the known–unknown boundary in the target domain. The execution flow for the AS module performing known–unknown separation on each batch of target domain samples is shown in Algorithm 1.

C. TA Module

Domain-level alignment and class-level alignment facilitate domain adaptation and knowledge transfer from two different perspectives. As shown in Fig. 4, domain-level alignment brings the overall distribution of shared classes between the source domain and target domain closer together, while class-level alignment further aligns the detailed distributions among the same classes within cross-domain shared classes. The following sections will introduce domain-level alignment and class-level alignment in the model.

1) Domain-Level Alignment: In the model, domain-level alignment is based on adversarial domain adaptation using a domain discriminator and a GRL. The features input to the domain discriminator are image patch features, which decompose the features of an image into patches that represent different regions of the image. The model employs a dual-head domain discriminator, which can simultaneously output prediction scores indicating whether the image patches belong to known classes or unknown classes [20]. Specifically, the model uses the dual-head domain discriminator and GRL to align the known class image patch features and unknown class image patch features of the two domains according to the adversarial domain adaptation methods outlined in the Section II. Due to the lack of clutter in source images, it is not possible to extract unknown class information from them. Therefore, when using L_{dca}, the model eliminates the alignment loss for unknown class image patch features in the target data with respect to the source unknown class. The domain-level alignment loss function L_{dca} used is as follows:

$$L_{\text{dca}} = -\sum_{i=1}^{|X_s|}\sum_{o}^{\{b,n\}} M_{o,s}^{i} D \log(f_o(x_s^i)) - \sum_{i=1}^{|X_t|}\sum_{o}^{\{b\}} M_{o,t}^{i} D \log(f_o(x_t^i)) \tag{10}$$

$$M_{b,s/t}^{i} = \text{Detach}\left(\sum_{k=1}^{K} p_c(y=k|x_{s/t}^i)\right)$$
$$M_{n,s/t}^{i} = \text{Detach}(p_c(y=K+1|x_{s/t}^i)) \tag{11}$$

where $|X_s|$ and $|X_t|$ represent the number of patch features in the source domain image features X_s and the target domain image features X_t. x_s^i and x_t^i denote the ith patch feature in $|X_s|$ and $|X_t|$. D is the domain label of the image, while $f_b(\cdot)$ and $f_n(\cdot)$ are the known class output head and unknown class output head of the dual-head domain discriminator G_d, respectively. $M_{b,s/t}^{i}$ and $M_{n,s/t}^{i}$ represent the known class mask and unknown class mask.

2) Class-Level Alignment: The prototypes learned through the prototype network can effectively represent the overall structure of each category, and the classification result of a single sample is obtained by calculating the distance between the projection of the sample in the embedding space and each prototype. Therefore, the model constructs known class prototypes separately in source–target domain spaces, and achieves accurate class-level alignment by reducing the distance between prototypes of the same category across domains. Given the absence of annotations in the target domain, the direct estimation of known-class prototypes within this domain is theoretically infeasible. To obtain reliable pseudoannotations for target domain samples, this work constructs a dynamically adaptive pseudolabel filtering framework governed by source-domain prototype supervision and a fixed threshold T_2, establishing optimal transport constraints across metric spaces. At the same time, by assigning different confidence scores to the acquired pseudolabels, we minimize the negative impact of potentially incorrect pseudolabeled target samples on the construction of target domain prototypes.

First, we learn the prototypes for the source domain using labeled source domain data. In the following equation, $P_s(y = Y_s^i|X_s^i)$ represents the label category score output by the source domain prototypes P_s for the source domain feature X_s^i in the embedding space

$$L_{ps} = \frac{1}{N_s}\sum_{i=1}^{N_s} -Y_s^i \cdot \log(P_s(y = Y_s^i|X_s^i)). \tag{12}$$

Then, the model employs supervisory guidance derived from source domain prototype representations, dynamically assigning pseudolabels to identified category instances within the target domain via fixed threshold T_2. Using source domain prototype supervision is because the constructed prototypes are the result of clustering, and the prototype-based classification method is distance-based. Therefore, the source domain prototypes contain clustering information that the classifier may lack in terms of extraction capabilities. This clustering information provides an additional discriminative perspective for both the selection of known class pseudolabels and the elimination of overlooked unknown class targets, thereby enhancing the accuracy of both processes. In addition, since there are only K known classes in the source domain, utilizing the source domain prototypes can also filter out unknown class samples, reducing redundancy when constructing target domain prototypes. Specifically, the filtering method first checks whether the predicted class of the classifier for the target sample is consistent with the predicted class of the source domain prototypes. If they are not consistent, no pseudolabel is assigned to that sample. Then, a fixed threshold T_2 is set to boost the reliability of the obtained pseudolabels Y_{tp}^i, meaning that a pseudolabel is assigned to the sample only if the maximum value in the one-hot prediction result is greater than the fixed threshold T_2.

$$L_{pt} = -\frac{1}{N_t'}\sum_{i=1}^{N_t'} Y_{tp}^i \cdot \log(P_t(y = Y_{tp}^i|X_t^i)) \tag{13}$$

where N_t' denotes the number of target samples with reliable pseudolabels. $P_t(y = Y_{tp}^i|X_t^i)$ represents the pseudolabel category score outputted by the target domain prototype for the target domain feature X_t^i in the embedding space.

To further ensure the accuracy of the constructed target domain prototypes, the impact of pseudolabel samples of varying quality on prototype construction should differ. For example, samples whose classifier predictions significantly exceed T_2 are regarded as high-confidence samples, and their influence on prototype learning should be enhanced; whereas samples that just exceed T_2 are considered low-confidence samples, and their influence on prototype learning should be diminished. Therefore, the model assigns different pseudolabel confidence C to corresponding samples based on the classifier prediction results of different samples, in order to reflect the differences in pseudolabel quality, and uses C as the loss weight for the corresponding samples in the training of target domain prototypes

$$C = \frac{\text{Max\_value} - T_2}{1 - T_2} \tag{14}$$

where Max_value is the maximum value in the softmax prediction of the classifier for target domain sample. The loss function for constructing target domain known class prototypes is defined as follows:

$$L_{pt\_c} = C * L_{pt}. \tag{15}$$

By optimizing the loss in (16), the model obtains prototypes of known classes for two domains

$$L_{\text{pro}} = L_{ps} + L_{pt\_c}. \tag{16}$$

$$L_{ca} = \frac{1}{K}\sum_{i=1}^{K} O(P_{si}, P_{yi}). \tag{17}$$

Finally, by utilizing the learned prototypes, we establish cross-domain prototype relationship—reducing the distance between known class prototypes of the same category across domains. Here, $O(\cdot)$ denotes the calculation of the Euclidean distance. By minimizing the distance loss in (17), we bring the distributions of the same category across domains closer together, achieving class-level alignment.

D. Overall Optimization Objective

In summary, during the training phase of the ASTA method, the joint optimization objective of the model is shown as follows:

$$L = L_s + L_{\text{fda}} + L_{\text{bp}} + L_{\text{dca}} + L_{\text{pro}} + L_{\text{ca}} \tag{18}$$

where L_{fda} is derived from the research by Li et al. [20]. L_{fda} aims to mitigate biased closed-set learning from the source domain by discovering and utilizing the hidden new class regions within the source domain images.

$$L_{\text{fda}} = \frac{|X_{lb}|}{|X_{lb} + X_n|} \cdot L_s + \frac{|X_n|}{|X_{lb} + X_n|} \cdot L_n \tag{19}$$

$$L_n = -\frac{1}{|X_n|}\sum_{x_i \in X_n} \log(p_c(y = K+1|x_i)) \tag{20}$$

where $|X_{lb}|$ and $|X_n|$, respectively, denote the number of labeled class image patch features and the number of newly mined class image patch features, while x_i represents the newly mined class image patch features in the image.

Specifically, the training process of the proposed ASTA method is as follows: The total number of training epochs is 155. During the first 20 epochs, we train the model using loss functions L_s, L_{fda}, and L_{ps} to establish preliminary classification capability for the K known classes in the source domain and to initialize prototypes for these K known classes in the source domain. From epoch 20 onward, we augment the training by incorporating loss functions L_{bp}, L_{dca}, L_{pt_c}, and L_{ca} in addition to the existing ones. In this phase: L_{bp} supervises the model to discriminate between known and unknown classes in the target domain, thereby developing known–unknown separation capability; L_{pt_c} establishes and dynamically updates prototypes for the K known classes in the target domain; L_{dca} and L_{ca} jointly facilitate cross-domain knowledge transfer through TA for known classes. During this training process, we consider all involved loss functions to be equally important, and therefore, assign equal weighting coefficients of 1 to each.

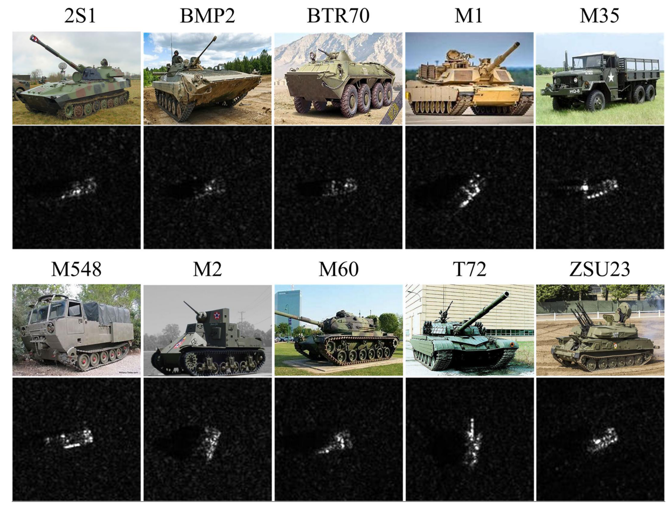

Fig. 5. Examples of ten target classes from the SAMPLE dataset. The SAMPLE dataset contains only ten classes of simulated and measured SAR images, without optical images; the optical images are merely illustrative examples.

IV. Experiments and Analysis

To rigorously examine the ASTA framework, this section conducts systematic experiments and multidimensional evaluations using the SAR dataset SAMPLE and optical dataset Office-31. The inclusion of the optical dataset serves to verify the generalization capability of the proposed ASTA method. First, the datasets and experimental settings are introduced, followed by comparisons with other OSDA methods. Then, analysis experiments are performed on various modules of the model, and the effect of hyperparameters on effectiveness is analyzed.

A. Experimental Setup and Implement Details

1) Dataset Description and Experimental Setup:

a) SAMPLE: The experiments in this article are primarily conducted on the SAMPLE dataset, which was made available in 2019 by the U.S. Air Force Research Laboratory [52]. This dataset comprises simulated and measured SAR images of various vehicle targets under different observation conditions, providing an excellent benchmark for OSDA recognition researching. The publicly available portion of the SAMPLE dataset includes ten types of ground military vehicle targets, as shown in Fig. 5, including 2S1 self-propelled howitzer, BMP2, BTR70 armored personnel carrier, M35, M548 truck, M1, M2, M60, T72 tank, and ZSU23 self-propelled antiaircraft gun. Table I presents the number of simulated and measured SAR data for each category, with azimuth angles ranging from 10°to 80°and elevation angles from 14°to 17°. In the experimental setup, the elevation angle for the test data is set at 17°, while the training data's elevation angles are 14°, 15°, and 16°, respectively.

TABLE I
DATA CATEGORIES AND CORRESPONDING QUANTITIES IN SAMPLE DATASET

Category	**2S1**	**BMP2**	**BTR70**	**M1**	**M35**	**M548**	**M2**	**M60**	**T72**	**ZSU23**
Simulation	174	107	92	129	129	128	128	176	108	174
Measured-Train	116	55	43	78	76	75	75	116	56	116
Measured-Test	58	52	49	51	53	53	53	60	52	58

In the subsequent experimental design, the domain containing simulated data is explicitly designated as the source domain (where complete class labels are accessible), while the domain containing measured data is established as the target domain. The first six categories of data in Table I are considered shared classes between the source and target domains, while the remaining four categories only appear in the target domain and are regarded as private classes. In addition, the target domain is divided into a training set and a testing set, with the training set participating in the learning of the cross-domain SAR open-set target recognition model alongside the source domain data. The testing dataset is utilized to evaluate the performance of the ASTA method.

b) Office-31: The Office-31 dataset is a widely used benchmark dataset in the field of visual transfer learning. It contains 31 common object categories in office environments, such as laptops, filing cabinets, and keyboards for image classification tasks, with a total of 4 652 images. The data in Office-31 is derived from three distinct domains (Amazon, Webcam, and DSLR), which exhibit significant differences in image quality, shooting angles, and backgrounds. In our experiments, following the standard experimental protocol for OSDA recognition methods, the source domain consists of the first ten classes (known classes), while the target domain includes the first ten classes (known classes) and the last 11 classes (unknown classes).

2) Evaluation Metrics: Following the evaluation metrics from mainstream OSDA research [14], [15], [16], [17], [18], [19], [20], [21], [22], this article adopts the same three evaluation metrics to assess model performance, including the average classification accuracy of known classes OS*, the rejection accuracy of unknown class UNK, and the harmonic mean of the two HOS as follows:

$$\text{OS}^* = \frac{1}{K}\sum_{i=1}^{K} A_i \tag{21}$$

$$\text{UNK} = \frac{n_1}{n} \tag{22}$$

$$\text{HOS} = 2 \cdot \frac{\text{UNK} \times \text{OS}^*}{\text{UNK} + \text{OS}^*} \tag{23}$$

where A_i represents the classification accuracy of the ith known class in the target domain. n denotes the total number of unknown class samples in the target domain, and n_1 indicates the number of samples in n that are correctly identified as belonging to the unknown class. HOS is the harmonic mean of OS* and UNK, and HOS is considered a core metric in OSDA recognition, as it requires the model to perform well on both known and unknown class.

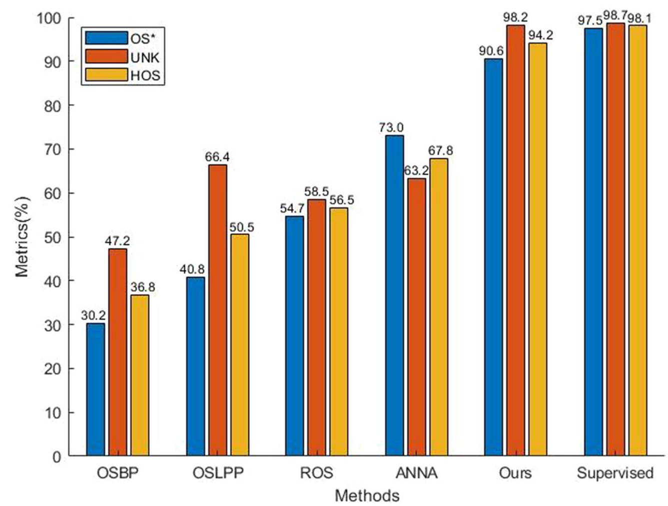

Fig. 6. Comparative efficacy (%) of strategies was quantitatively assessed under the OSDA recognition implemented with the SAMPLE.

3) Implement Details: Considering that the information in SAR images is relatively simple, in order to avoid overfitting due to a too deep network, the model uses ResNet18 as the feature extractor. Before the formal training of the model, we set up a warm-up pretraining phase (the first 20 epochs), using only the source domain data to train the feature extractor, classifier, and source domain prototypes to help the model better adapt to the data. The experimental framework employs a stochastic gradient descent optimizer architecture for model training, with the base learning rate configured at 0.001 (1e-3), batch size is 32, momentum factor calibrated to 0.9, and training cycle ceiling constrained to 155 epochs. The hyperparameters T_1, T_2, and α in the method are set to 0.33, 0.62, and 3.5, respectively.

B. Comparative Experiments

1) Overall Performance: To assess the performance of the ASTA method proposed in this study, a systematic comparison was made with the results of typical OSDA identification methods. In Fig. 6, we demonstrate the comparative outcomes of the OSDA recognition task on SAMPLE dataset. OSBP [15], OSLPP [19], ROS [17], and ANNA [20] are several typical OSDA methods, with ANNA serving as the baseline model for the ablation experiments in this article. The “Supervised” indicates the model training using fully labeled data from the measured SAR domain and testing on the measured SAR data domain, representing the upper bound for the OSDA recognition task on the SAMPLE dataset. These OSDA comparison

TABLE II
STATISTICAL RESULTS (%) OF CROSS-DOMAIN OPEN-SET EXPERIMENTS FOR DIFFERENT OSDA RECOGNITION METHODS ON THE OFFICE-31 DATASET

Method	A→D			A→W			D→A			D→W			W→A			W→D			Average		
	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS
OSBP [15]	90.5	75.5	82.4	86.8	79.2	82.7	76.1	72.3	75.1	97.7	96.7	97.2	73.0	74.4	73.7	99.1	84.2	91.1	87.2	80.4	83.7
OSLPP [19]	92.6	**90.4**	**91.5**	**89.5**	**88.4**	**89.0**	**82.1**	76.6	79.3	96.9	88.0	92.3	**78.9**	78.5	78.7	95.8	91.5	93.6	**89.3**	85.6	87.4
ROS [17]	87.5	77.8	82.4	88.4	76.7	82.1	74.8	81.2	77.9	99.3	93.0	96.0	69.7	86.6	77.2	**100.0**	**99.4**	**99.7**	86.6	85.8	85.9
ANNA [20]	**93.2**	76.1	83.8	82.8	**88.4**	85.5	75.4	91.1	82.5	**99.4**	**99.6**	**99.5**	76.0	87.9	81.6	**100.0**	96.8	98.4	87.8	90.0	88.6
Ours	89.8	85.3	87.5	87.2	88.1	87.6	77.2	**92.0**	**83.9**	98.2	96.3	97.2	77.1	**88.1**	**82.2**	97.3	96.9	97.1	87.8	**91.1**	**89.3**

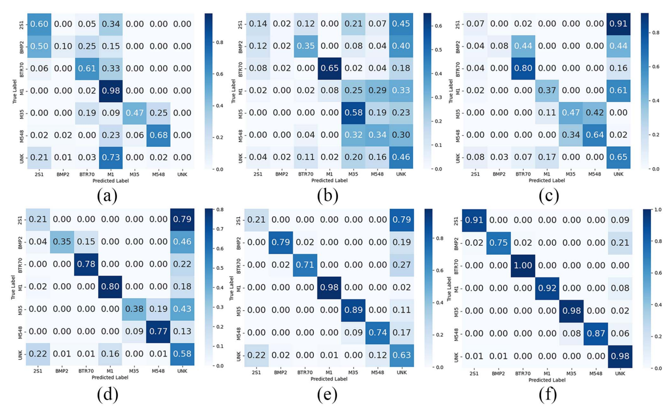

Fig. 7. Confusion matrices (%) of different methods on the SAMPLE dataset. (a) Source. (b) OSBP. (c) OSLPP. (d) ROS. (e) ANNA. (f) Ours.

methods improve cross-domain open-set classification effects by introducing transfer learning strategies and open-set recognition strategies, highlighting the varying effects of different strategies in OSDA. The proposed method achieves the best results among all OSDA recognition methods. Compared to the baseline model ANNA, ASTA optimizes the separation process through the AS mechanism, and simultaneously, utilizes TA mechanism to achieve feature alignment at two levels, resulting in significant improvements. Notably, the performance of ASTA is nearly comparable to fully supervised classification methods.

To validate the generalization capability and robustness of the proposed ASTA method, we conducted comparative experiments on Office-31, a benchmark dataset widely used in OSDA recognition. The experimental results are presented in Table II. As shown in the results, our method achieves the highest HOS scores in D→ A and W→ A tasks. Across all six transfer tasks, the proposed method attains an average HOS of 89.3% . These experimental results demonstrate that our approach exhibits promising generalization capability and robustness.

2) Confusion Matrix: In classification tasks, the confusion matrix is commonly used to demonstrate the classification performance of target recognition models. It clearly presents the classification results for each category in a matrix format. The true classes are represented by the rows of the confusion matrix, while the predicted classes are indicated by the columns. Each element in the confusion matrix shows the probability of an object being classified into a particular class, with the diagonal elements reflecting the classification accuracy for that class.

Fig. 7 shows the confusion matrices for different methods performing the OSDA recognition task on the SAMPLE dataset. The "Source" method in Fig. 7(a) involves training the model using simulated SAR data, and then, directly testing it on the measured SAR data domain. Fig. 7(a) indicates that there is a significant domain shift issue between simulated SAR data and measured SAR data, and the model trained solely with shared class data from the simulated SAR domain lacks open-set recognition capability on the measured SAR data domain, resulting in poor performance of the Source model on measured SAR data. Several OSDA comparison methods address domain shift and open-set issues to some extent by employing different transfer learning strategies and open-set recognition strategies. However, these OSDA methods still have considerable room for improvement. Notably, the proposed method implements AS mechanism, allowing the model to extract more discriminative feature representations between known and unknown classes in the target domain. In addition, through TA mechanism, the model can extract DIR for known classes across domains that are less separable, while also extracting more distinguishable features representation among known classes, facilitate the model's ability to precisely align the distribution of shared classes across domains. Ultimately, the model achieves great classification results.

3) Feature Visualization: In order to demonstrate the separation and cross-domain alignment effects of the proposed method from a feature perspective, Fig. 8 presents the t-SNE feature visualizations of five comparison methods and the proposed method ASTA. In Fig. 8, the different colors of the sample points represent different classes, where the light-colored series of sample points indicates the feature projection of simulated SAR data, the dark-colored series represents the feature projection of measured SAR data, and the gray represents the feature projection of unknown class data from the measured SAR domain.

Fig. 8(a) shows the t-SNE feature visualization results of the Source method tested on measured SAR data, as described in the previous subsection. From a feature perspective, it can be observed that there is a significant domain gap between the data distributions of the simulated SAR domain and the measured SAR domain. From Fig. 8(b)–(e), it can be observed that OSBP, OSLPP, and ANNA have separated known and unknown class features to some extent in the measured SAR domain by introducing different open-set operations and domain adaptation losses, while also reducing the domain gap between the feature distributions of shared class features in simulated SAR data and measured SAR data. Compared to Fig. 8(a), OSBP, OSLPP,

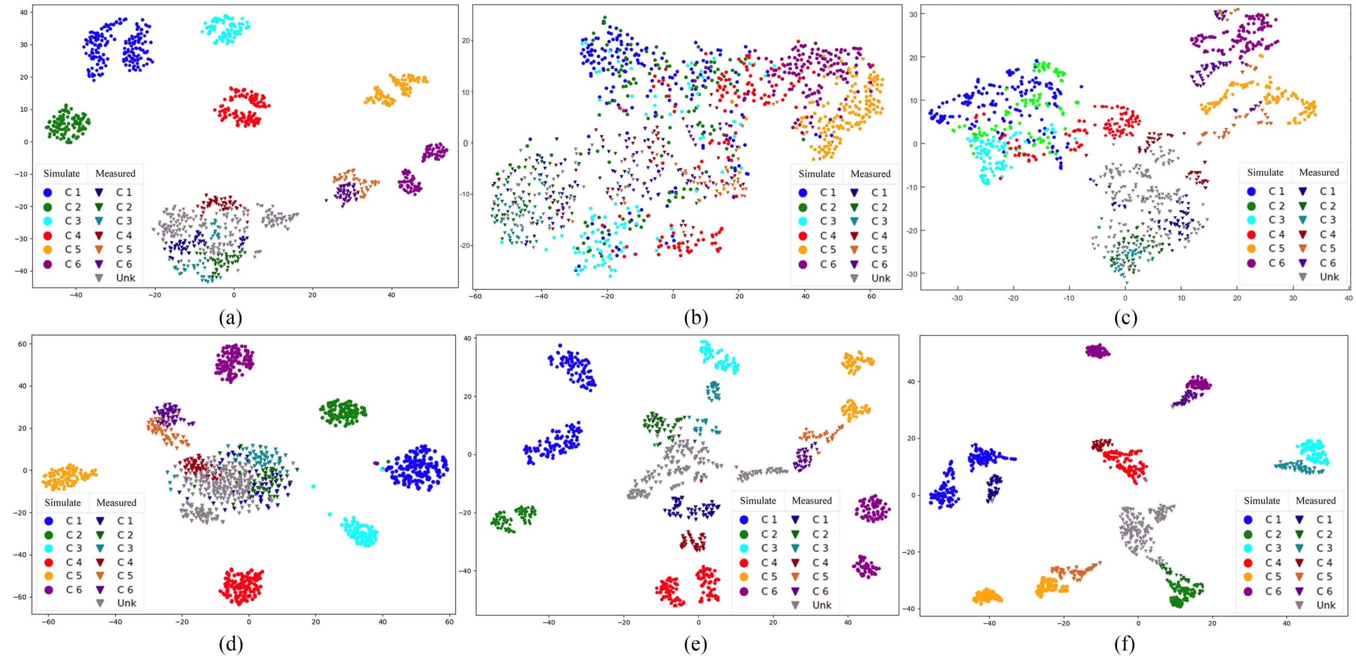

Fig. 8. Comparison of t-SNE feature visualization results for the (a) Source, (b) OSBP, (c) OSLPP, (d) ROS, (e) baseline model ANNA, and (f) proposed method ASTA under the simulate SAR → measured SAR OSDA recognition task. C1–C6 represent 2S1, BMP2, BTR70, M1, M35, and M548, respectively, and Unk represents the unknown class in the measured SAR domain.

and ANNA have had varying degrees of negative impact on the class structure in the simulated SAR domain–leading to reduced intraclass feature compactness and increased interclass feature dispersion. However, they have succeeded to some extent in separating known and unknown features in the measured SAR domain and enhancing the similarity of known class features between the measured SAR domain and the simulated SAR domain. However, from a feature perspective, it is evident that the performances of OSBP, SOLPP, and ANNA in terms of separation and alignment are still not ideal. The ROS t-SNE plot shown in Fig. 8(d) differs from the other three OSDA comparison methods. This is because ROS employs an adjusted rotation recognition task, which does not significantly impact the sample features. The t-SNE feature visualization results of the ASTA are shown in Fig. 8(f). The ASTA results in better feature dispersion between known and unknown classes in the measured SAR domain, reduces feature distribution differences among shared classes across domains.

4) Model Complexity and Efficiency: We conducted comprehensive comparisons between the proposed method and existing approaches in terms of training time (TrainT), inference time (TestT), parameter count (Para), and FLOPs to evaluate model complexity and efficiency, thereby demonstrating the practical significance of our method. The statistical results are presented in Table III. Among the compared methods, OSLPP is a traditional (nondeep learning) approach, and thus, excluded from complexity and efficiency analysis since it is not a trainable deep learning model. The other four methods are all deep learning-based. The statistics in Table III are based on the SAMPLE dataset under the experimental results shown in Fig. 6, following the division of six known classes and four unknown classes in the target domain, with all time metrics representing multiepoch averages. As shown in Table III, the proposed method achieves: Relatively slower training time (attributed to the two proposed modules); second-best inference time performance; and the lowest parameter count and FLOPs among all compared methods.

TABLE III
STATISTICAL RESULTS OF COMPLEXITY AND EFFICIENCY METRICS FOR DIFFERENT OSDA RECOGNITION METHODS ON THE SAMPLE

Method	TrainT(s)	TestT(s)	Para(M)	FLOPs(G)
OSBP	3.331	0.624	28.57	4.13
ROS	18.265	3.503	12.08	7.29
ANNA	3.183	0.941	11.77	1.82
Ours	3.672	0.941	11.77	1.82

5) Openness Degree Analysis: Under open-set conditions, open-set recognition methods typically validate robustness across different known/unknown class partitioning ratios (i.e., varying openness degrees) [58], [59], [60]. Inspired by this, we conducted robustness verification of different OSDA methods under varying openness degrees on the SAMPLE dataset. The experimental results are presented in Table IV, where "30% " indicates that unknown classes constitute 30% of the total categories (i.e., an openness degree of 30%). As evidenced by the results in Table IV, the proposed method consistently achieves optimal performance compared to existing OSDA approaches across varying openness degrees on the SAMPLE dataset, demonstrating its superiority for SAR data processing. It is observed that the performance of all methods degrades to a certain extent as the openness increases. This is because with the

TABLE IV
EXPERIMENTAL RESULTS (%) OF DIFFERENT OSDA METHODS UNDER VARYING OPENNESS (%) DEGREES ON THE SAMPLE DATASET

Method	30%			40%			50%			60%		
	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS	OS*	UNK	HOS
OSBP	27.6	63.3	38.4	30.2	47.2	36.8	34.7	28.3	34.8	38.0	17.1	29.3
OSLPP	37.6	61.2	46.6	40.8	66.4	50.5	39.2	60.5	47.6	39.9	59.9	47.9
ROS	15.1	56.2	23.8	54.7	58.5	56.5	49.0	58.4	53.3	28.1	46.4	35.0
ANNA	71.3	94.1	81.2	73.0	63.2	67.8	59.9	63.0	61.4	67.6	64.4	66.0
Ours	**90.0**	**95.2**	**92.5**	**90.6**	**98.2**	**94.2**	**79.4**	**82.6**	**81.0**	**84.9**	**72.9**	**78.5**

TABLE V
ABLATION EXPERIMENT RESULTS (%) OF THE PROPOSED METHOD ASTA ON SAMPLE DATASET

Row	AS		DA	CA		OS*	UNK	HOS
	Without ω	**With ω**		**Without C**	**With C**			
1						73.0	63.2	67.8
2	✓					70.0	72.2	71.1
3	✓		✓			70.1	74.9	72.4
4	✓		✓	✓		76.7	94.2	84.7
5	✓		✓		✓	82.3	93.7	87.6
6		✓	✓		✓	**90.6**	**98.2**	**94.2**

TABLE VI
STATISTICAL RESULTS OF COMPLEXITY AND EFFICIENCY METRICS FOR DIFFERENT OSDA RECOGNITION METHODS ON THE SAMPLE

Row	DA	AS	CA	Train-Time(s)	Test-Time(s)
1				3.183	0.941
2	✓			3.182	0.941
3	✓	✓		3.534	0.941
4	✓	✓	✓	3.672	0.941

TABLE VII
STATISTICS OF SEPARATION PSEUDOLABEL ACCURACY (%) FOR FIXED-THRESHOLD AND AS MODULE

Separation Ways	Known-class Correct num	Unknown-class Correct num	Average Accuracy
Fix-th 0.5	262	197	85.6
AS	286	219	94.4

increase in openness, the proportion of unknown classes in the target domain rises accordingly, while the amount of labeled data in the source domain decreases. As a result, the contradiction between the model's reliance on the known classes in the source domain and the uncertainty of the unknown classes in the target domain is amplified. Moreover, the feature distribution of the unknown classes in the target domain will cover the feature space more extensively. Both the complexity of their feature distribution and the degree of overlap with the features of the known classes in the target domain will increase the difficulty for the model to separate the known and unknown classes in the target domain and align the known classes across domains.

C. Model Analysis

1) Ablation Study: In order to analyze the impact of different modules and improvements of ASTA on SAR OSDA recognition performance, this article presents ablation experiments from the baseline model to the complete model in Table V, where DA and CA represents the domain-level alignment submodule and class-level alignment submodule in the TA module, respectively. Compared to the baseline model in Row 1, Row 2 shows that changing the fixed threshold in the baseline model to an adaptive threshold ρ allows each batch of target samples to obtain a suitable separation threshold for known and unknown classes in the target domain. The adaptive threshold ρ is more reasonable than the fixed threshold, leading to a 3.3% increase in HOS, which also demonstrates the superiority of the adaptive threshold ρ over the fixed threshold. Next, the model employs the adjusted domain-level alignment (Row 3), which brings a 1.3% improvement in HOS, validating that our adjustments are reasonable and effective. Building on the coarse separation and domain-level alignment, Row 5 introduces a class-level alignment submodule with pseudolabel confidence, significantly enhancing the alignment effect for known classes and the rejection accuracy for unknown class samples, showcasing the substantial benefits of class-level alignment for the model. The comparison between Row 4 and Row 5 also illustrates the role of the pseudolabel confidence C. Finally, Row 6 incorporates target sample separation weights ω into the progressive AS module, forming a complete method that achieves 90.6% OS*, 98.2% UNK, and 94.2% HOS on the SAMPLE dataset.

To analyze the time impact of each proposed module on model training and testing, as shown in Table VI, we examined the

TABLE VIII
STATISTICS OF PSEUDOLABELING SITUATIONS AND TEST RESULTS (%) FOR DIFFERENT PSEUDOLABEL FILTERING METHODS IN CLASS-LEVEL ALIGNMENT SUBMODULE

Filering Ways	Selected Num	Known-class Num	Known-class Correct Num	Pseudo-label Cor-Accuracy	OS*	UNK	HOS
Fixed-th T_2	504	353	270	76.5	81.2	94.2	87.2
PS + Fixed-th T_2	323	323	261	80.8	90.6	98.2	94.2

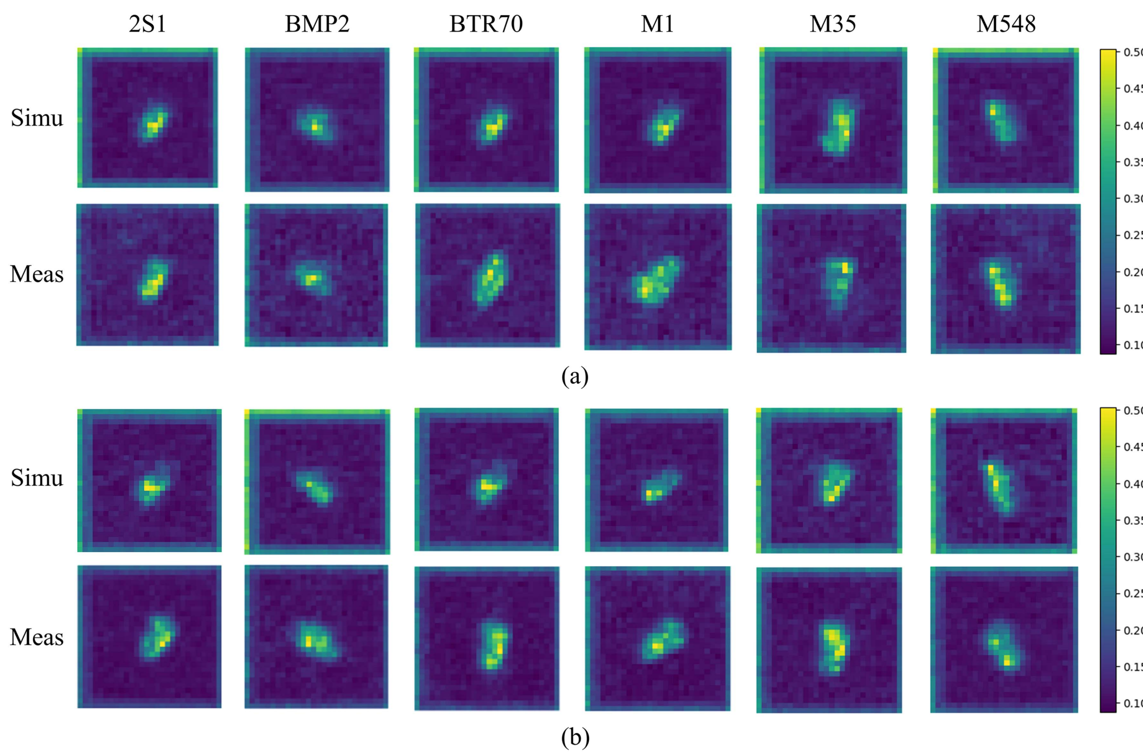

Fig. 9. Typical examples of low-level feature map visualization for the TA effects. The varying brightness in different areas of a feature map indicates the level of attention the network pays to each area: the higher the brightness, the greater the attention. The first and second rows represent the feature maps corresponding to different categories learned by the feature extraction network from the original simulated/measured SAR images before TA. The third and fourth rows represent the feature maps corresponding to different categories learned by the feature extraction network from the original simulated/measured SAR images after TA. (a) Before two-level alignment. (b) After two-level alignment.

incremental time effects of progressively adding the proposed modules to the baseline model. The time reported in Table VI represents the average time across all epochs. From Row 2, it can be observed that modifying the domain alignment loss function in the baseline model did not significantly affect training time. Row 3 demonstrates that after incorporating the AS module, the training time increased by 0.351 s (11.03%) compared to the baseline model. This is because each batch requires the computation of the adaptive threshold ρ and the separation loss weights ω for all samples of target domain in the batch. As seen in Row 4, further adding the class-level alignment module increased the training time by 0.138 s (3.90%) relative to Row 3. This overhead stems from constructing known-class prototypes for both source and target domains per batch, along with calculating the construction loss weights C for target domain samples involved in prototype building. The ASTA model exhibited a total training time increase of 0.489 s (15.36%) compared to the baseline model. Since the test models saved under all four models consist of the trained feature extractor and classifier, their inference times are identical.

2) AS Separation Effect: In order to intuitively compare the different effects of fixed threshold separation (Fixed-th 0.5) and AS module, this article conducts a statistical analysis of the separation results of these two methods on the test data, as shown in Table VII. The analysis includes the number of correctly separated known class target samples, the number of correctly separated unknown class target samples, and the average accuracy for both. In terms of quantity, the AS mechanism leads by 24 known classes samples and 22 unknown class samples, compared to the fixed threshold. In terms of proportion, the AS mechanism has a lead of 7.6% and 9.9% for the known and unknown classes, respectively. In the decision boundary delineation between known and unknown classes in the target domain, the AS mechanism successfully separates additional samples that the fixed threshold fails to classify correctly. The separation results indicate that the AS mechanism can delineate more refined known and unknown decision boundary between. The advantages of the AS module not only directly contribute to improving the rejection accuracy of unknown class within the target domain, but also indirectly promote the correct alignment of shared class features across domains during the alignment phase.

3) Pseudolabel Filtering Effect: The performance of prototype-based class-level alignment methods are often related to the predicted pseudolabels. Without sufficient filtering mechanisms, incorrect pseudolabels can negatively impact class-level alignment, leading to the entanglement between known and unknown classes, as well as negative alignment among known class intraclass categories. In this article, ASTA utilizes the predictions of source domain prototypes (PS) to filter the target samples in the prototype class-level alignment submodule. Compared to classifier, source domain prototypes are constructed based on clustering principles, thus possessing stronger clustering information. In addition, since the source domain prototypes are supervised by the categories of source domain data, they can inherently filter out unknown class target samples with pseudolabels, thereby reducing redundancy in constructing target domain prototypes.

To rigorously assess the efficacy of the newly developed pseudolabel filtering mechanism, this article compares the pseudolabel filtering situations using only a fixed threshold versus using both source domain prototype training supervision and a fixed threshold on the measured domain training data of the SAMPLE dataset, as well as the impact of these two pseudolabel filtering methods on the model's test results on the measured SAR test data. As shown in Table VIII, the method of filtering pseudolabels using only a fixed threshold resulted in a greater number of target samples for constructing target domain prototypes, but it failed to prevent unknown class pseudolabel samples from forming unknown class target domain prototypes. This approach also generated more incorrect pseudolabels for known classes, hindering the accurate learning of target domain known class prototypes and subsequently harming the overall accuracy of the model. The proposed pseudolabel filtering method (PS + Fixed-th T_2) first avoids establishing target domain unknown class prototypes, reducing redundancy in the model, and generates fewer incorrect pseudolabels for known classes, ensuring the correct guidance of class-level alignment, which in turn improves the accuracy of the model.

4) Feature Map Alignment Effect: In convolutional neural networks, feature maps are spatial representations of the features extracted by the network from the input data, carrying the feature information extracted at different levels. While high-level feature maps capture more complex structures and semantic information, resulting in a higher degree of abstraction, low-level feature maps better reflect the structural and contour information of objects within the images. Therefore, to further substantiate the efficacy of the two-layer alignment approach presented in this article, we have elected to depict the low-level feature maps derived from both synthetic and measured SAR imagery. This visualization aims to clearly illustrate the transformations imparted to the low-level feature maps by the model, both before and after the implementation of the two-layer alignment module.

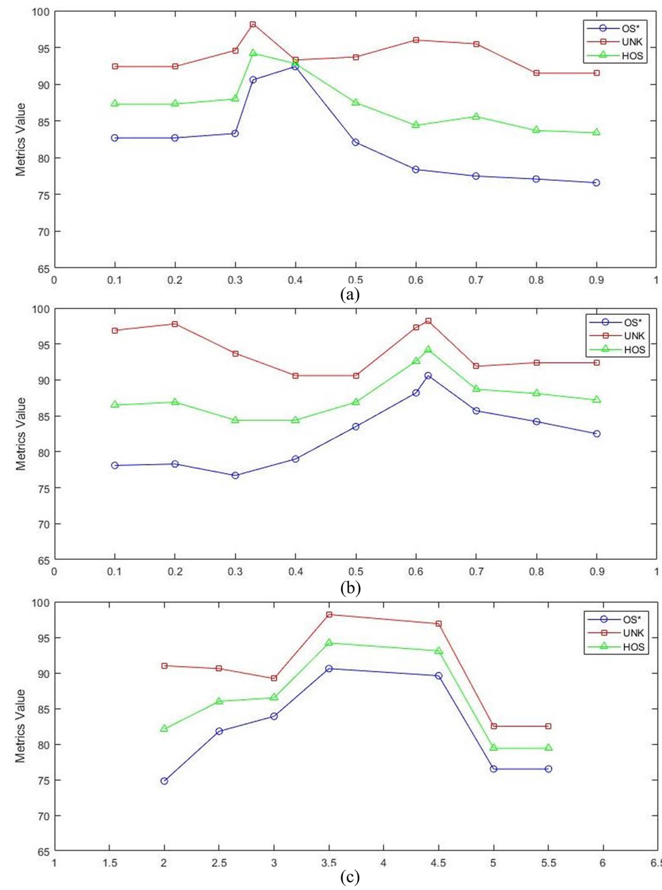

Fig. 10. Hyperparameter analysis experiment (%). This figure presents the analysis of hyperparameters, including (a) threshold T_1 that controls the participation of target samples in distance calculations within the AS module, (b) threshold T_2 used to filter pseudolabels of target samples in class-level alignment submodule, and (c) amplification factor α representing the ratio of the minimum distance m_1 between classification centers and classification loss m_2 of each batch of target samples.

Fig. 9 displays the low-level feature maps of simulated SAR images and measured SAR images before and after the implementation of the TA module. From Fig. 9(a), it can be observed that before the TA, there are significant differences in the feature maps of simulated SAR images and measured SAR images; the feature map of the simulated SAR image is relatively clean with less noise, while the feature map of the measured SAR image is quite noisy. From Fig. 9(b), it can be observed that after the TA, the low-level feature map of the simulated SAR image tends to become noisier, whereas the measured SAR image tends to be denoised, and the features of the simulated and measured domains become increasingly confused. The differences between before TA and after TA reflect the alignment effect of the TA module, validating the effect of the TA module from the perspective of low-level features.

5) Hyperparameter Analysis: This part analyzes the hyperparameters in ASTA, as shown in Fig. 10. As illustrated in Fig. 10(a), different values of T_1 directly affect the delineation of the decision boundary between known and unknown classes in the target domain, thereby influencing the overall model. Initially, as T_1 gradually increases, a portion of incorrect samples is removed, resulting in fewer erroneous samples contributing to the calculation of the adaptive threshold ρ. At this point, ρ accurately reflects the distance relationship between known class samples in the batch and their respective classification centers. However, as T_1 becomes excessively large, not only are incorrect samples discarded, but an increasing number of correctly classified known class samples with low predicted scores are also removed, which leads to ρ being unable to adequately reflect the distance relationship among known class samples within the batch, negatively affecting overall performance. As illustrated in Fig. 10(b), we investigated the impact of different values of T_2 on the construction of known class prototypes in the target domain and their overall performance. As T_2 gradually increases, some incorrectly labeled target samples are excluded, facilitating the accurate construction of prototypes, which enhances the domain alignment precision for known classes and positively influences the delineation of the decision boundary between known and unknown classes. However, when T_2 becomes excessively large, some correctly labeled known class samples with insufficient predicted scores may be excluded, leading to an insufficient number of pseudolabeled known class samples in the target domain. Consequently, the constructed prototypes may not adequately represent the structure of the corresponding classes, adversely affecting the alignment of known classes. The choice of α is related to the dataset, with different datasets having different values for α. As shown in Fig. 10(c), the optimal choice of α for the sample dataset is 3.5.

V. Conclusion

To address the issues of imprecise separation and negative alignment in existing OSDA methods, this article proposes an OSDA recognition framework called ASTA, which effectively minimizes the influence of unknown classes during knowledge transfer, thereby achieving accurate classification of known classes and precise identification of unknown classes in the target domain. The proposed AS module enables precise delineation of the boundary between known and unknown classes in the target domain through coarse-to-fine separation. By introducing the multilevel alignment concept from CSDA into OSDA recognition, the proposed TA module simultaneously aligns both the overall distribution and individual class distributions of shared categories at domain-level and class-level, facilitating effective and accurate knowledge transfer from source to target domain. Finally, the outstanding performance in OSDA recognition tasks from simulated to measured SAR data comprehensively validates the effectiveness and superiority of the ASTA framework.

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